

NITRITE

Nitrite nitrogen occurs as an intermediate stage in the biological decomposition of compounds containing organic nitrogen. Nitrites are not often found in surface waters because in aerobic conditions they are readily oxidized to nitrates. Levels of nitrites greater than natural residual amounts can be acutely toxic to fish. The test shown below is very sensitive to low nitrite concentrations.

Procedure

1. Rinse a sample tube several times with the sample, then fill to the 5-mL mark. Add the contents of one NitriVer®* 3 Powder Pillow for 5-mL Sample.
2. Stopper the tube and shake vigorously for exactly one minute. A red color will develop if nitrite is present.
3. Place the Nitrite Nitrogen Color Disc into the Color Comparator if it is not already in place.
4. Allow the prepared sample to sit undisturbed for 10 minutes, then place the tube into the right-hand opening of the comparator. (See *Figure 1* on page 8.)
5. Fill the other viewing tube to the 5-mL mark with the original sample water. Stopper and place it into the left-hand opening.
6. Hold the Comparator up to a window, the sky, or a lamp and view through the openings in front. Rotate the disc until a color match is obtained.
7. Read the mg/L nitrite nitrogen (N) through the scale window.

Note: The test results can be converted from mg/L nitrite nitrogen (N) to mg/L nitrite (NO_2^-) by multiplying the final reading by 3.3.

REAGENTS AND APPARATUS

Description	Cat. No.
Color Comparator Box.....	1732-00
Color Disc, nitrite nitrogen	14084-00
Color Viewing Tubes, plastic (4)	46600-04
NitriVer® 3 Powder Pillows (100)	14078-99

* NitriVer is a registered trademark of Hach Company.